Title: Car Price Prediction Using Machine Learning

Abstract

This project focuses on predicting used car prices using machine learning techniques. The dataset contains important features such as car brand, car model, fuel type, kilometers driven, city, and manufacturing year. Data preprocessing and exploratory data analysis (EDA) were performed to clean and understand the dataset. A Random Forest Regressor model was used for prediction, achieving a high accuracy with an R^2 score of approximately 0.94. This system can help buyers and sellers estimate fair car prices in real-world scenarios.

Introduction

Used car price prediction is an important problem in the automobile industry. The price of a car depends on multiple factors such as brand, age, mileage, and fuel type, making it difficult to estimate manually.

With the growth of online car marketplaces, predicting accurate car prices has become essential for both buyers and sellers. Machine learning provides a powerful approach to analyze historical data and build models that can predict prices efficiently and accurately.

Problem Statement

The objective of this project is to develop a machine learning model that predicts the price of a used car based on features such as:

- Brand
- Car Model
- Fuel Type
- Kilometers Driven
- City
- Car Age

📊 Dataset

- **Source:** Used Car Dataset
- **Number of Records:** (Add your dataset size here, e.g., 5000+)

Features:

- `kms_driven` (Numerical)
- `car_age` (Numerical)
- `brand` (Categorical)
- `car_model` (Categorical)
- `fuel_type` (Categorical)
- `city` (Categorical)
- `price` (Target Variable)

⚙️ Data Preprocessing

The dataset was cleaned and prepared using the following steps:

- Removed missing values to ensure data quality
- Converted the `price` column to numeric format
- Split the car name into two separate features: **brand** and **car model** to improve model performance
- Applied **Label Encoding** for brand, car model, and fuel type
- Applied **One-Hot Encoding** for city
- Detected and removed outliers to improve prediction accuracy
- Created a new feature called **car_age** from the manufacturing year

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☑ Exploratory Data Analysis (EDA)

EDA was performed to understand patterns and relationships in the dataset:

- Visualized the distribution of car prices
- Observed that **newer cars tend to have higher prices**
- Identified that **premium brands have higher resale value**
- Found relationships between:
  - Mileage and price
  - Fuel type and price
- Detected outliers in price and mileage

🤘 Model Building

- **Model Used:** Random Forest Regressor

◈ Why Random Forest?

- Handles both categorical and numerical data
- Captures non-linear relationships effectively
- Reduces overfitting using multiple decision trees
- Provides high prediction accuracy

📏 Model Evaluation

Performance Metrics:

- **Train R^2 Score:** 0.95
- **Test R^2 Score:** 0.94
- **MAE (Mean Absolute Error):** ~100,000
- **RMSE (Root Mean Squared Error):** ~170,000

📌 Interpretation:

- R^2 score close to 1 indicates excellent model performance
- MAE shows the average prediction error
- RMSE highlights larger errors and ensures model reliability

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📊 Results

- The model successfully predicts used car prices with high accuracy
- Important features influencing price include:
  - Car brand
  - Car age
  - Kilometers driven
- The model shows good generalization and does not overfit

✅ Conclusion

This project demonstrates that machine learning can effectively predict used car prices. The Random Forest model performed well with high accuracy and reliable predictions. This approach can be applied in real-world applications such as online car marketplaces.

🚀 Future Work

- Increase dataset size for better accuracy
- Try advanced models like XGBoost or Gradient Boosting
- Build a web application using Streamlit
- Deploy the model for real-time predictions

📚 References

- Python Libraries:
  - Pandas
  - NumPy
  - Scikit-learn
  - Matplotlib
  - Seaborn
  - Joblib
  - SciPy
- Dataset Source: Used Car Dataset